

- 7 It has been hypothesised that the mutation rate of an animal species may affect how fast animals of that species age and how long they live (lifespan).

Table 7.1 compares the mutation rate and lifespan of five species of mammal.

Table 7.1

species	mutation rate per million base pairs per year	mean lifespan /years
cattle	0.08	20
ferret	0.20	8
horse	0.05	30
human	0.02	75
house mouse	0.28	3.6

- (a) (i) Describe the relationship shown in Table 7.1.

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- (ii) Use Table 7.1 to explain how these species may differ in the rate at which they can adapt if environmental selection pressures change.

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- (b) (i) Compare the timescales needed for selective breeding and genetic engineering to increase the frequency of a rare allele in cattle. Give a reason for your answer.

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- (ii) Genetic engineering could be used to improve farmed animals, such as dairy cattle.

State **two** features of dairy cattle that could be improved by genetic engineering.

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[Total: 8]